

CourseHub API - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

CourseHub API is a RESTful learning platform enabling instructors to create structured courses and students to manage enrollments. The system implements a shared identity model with role-based access control to isolate content ownership and enrollment data. Core value is delivered through a strict course-module hierarchy and progress tracking mechanisms hosted on a v1 API.

1.2 Maturity Assessment

The specification is technically robust and structurally sound, characterized by clear success criteria and precise edge-case definitions. While a few medium and low severity gaps exist regarding the formal definition of out-of-scope boundaries and minor uncertainties, they do not impede the current technical trajectory. Status: READY.

1.3 Technical Stack

- REST
- JSON
- Resend

1.4 Architectural Constraints

- API endpoints must be hosted under `/api/v1/`.
- JSON response envelope requirement: `{"data": ..., "meta": ..., "errors": []}`.
- Course prices must be stored strictly in cents.
- Enrollment progress must be a value between 0 and 100 inclusive.
- Course modules must maintain a fixed sequence ordering.
- Core course, enrollment, and authorization behaviors must have $\geq 90\%$ E2E test coverage.
- Instructors can only manage/delete courses they own.
- Students can only access and update their own enrollments.
- Enrollment is strictly forbidden for unpublished courses.

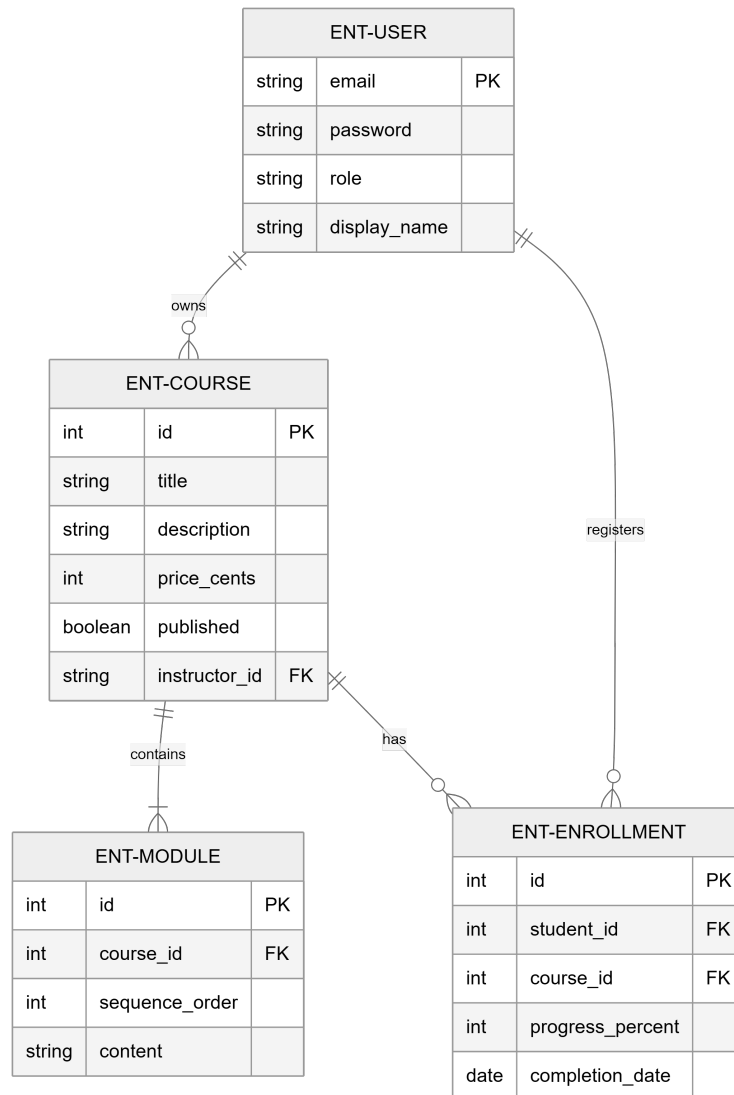
1.5 Critical Dependencies

- `RESEND_API_KEY` environment variable for welcome email dispatch.
- Strict foreign key dependence between Enrollment, User, and Course entities.
- Course deletion blocked by existence of active Enrollment records.
- Shared User table utilizing a 'role' field for access control differentiation.
- Course-to-Module one-to-many relationship with sequence preservation.

2. Architecture Workflows & Visual Diagrams

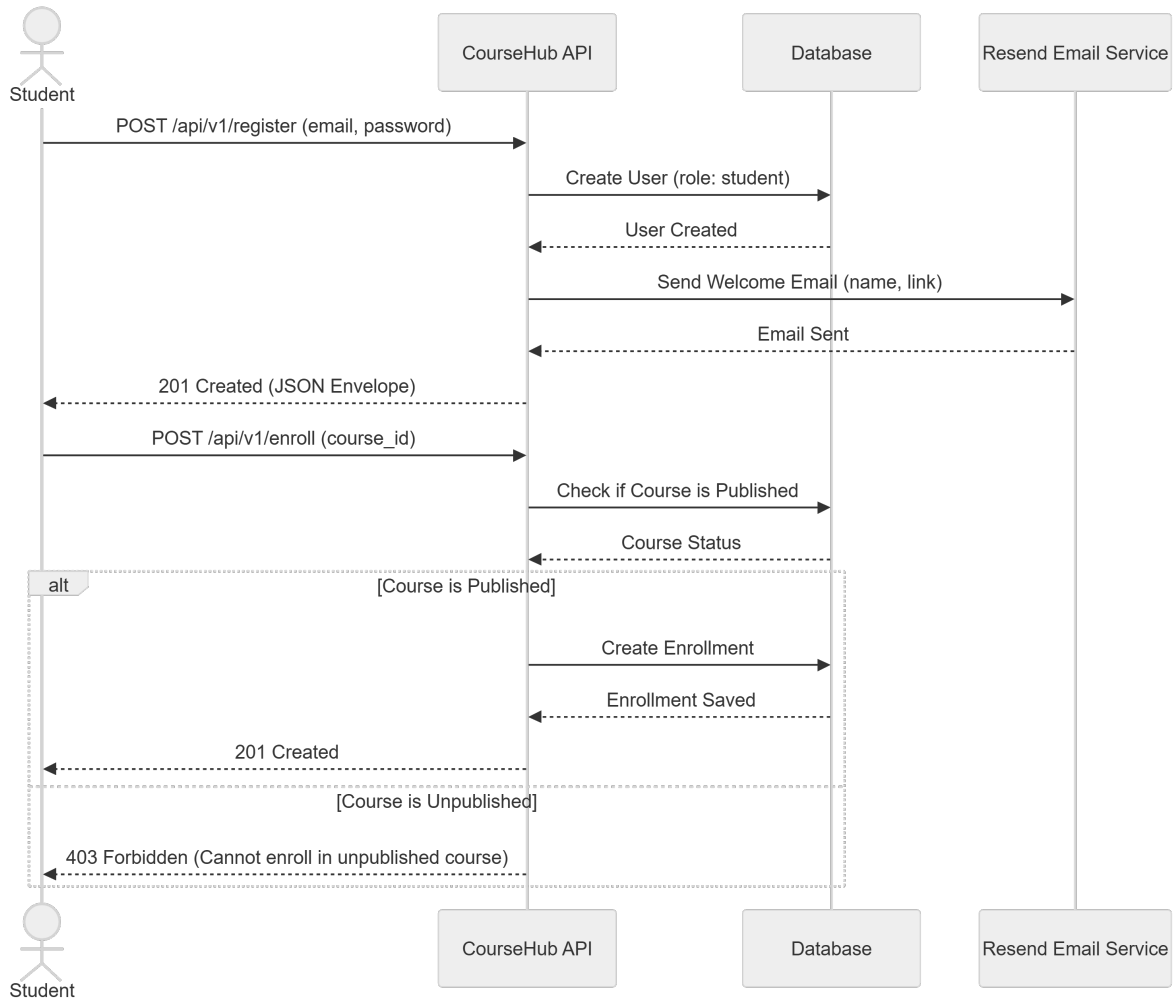
2.1 CourseHub Data Model

Entity Relationship Diagram showing the shared user table, course ownership, module sequencing, and student enrollments.



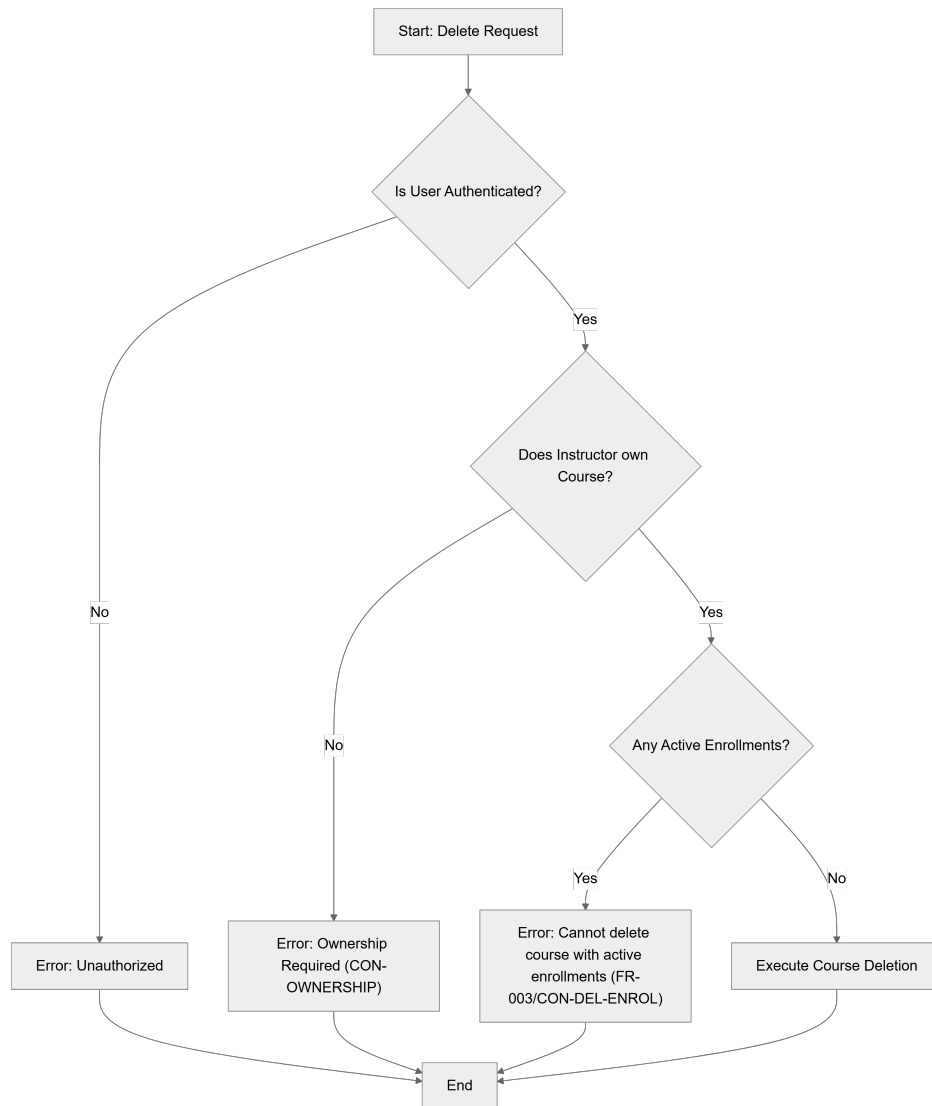
2.2 Student Registration & Enrollment Flow

Sequence of interactions for a student registering and enrolling in a course, including the external Resend email integration.



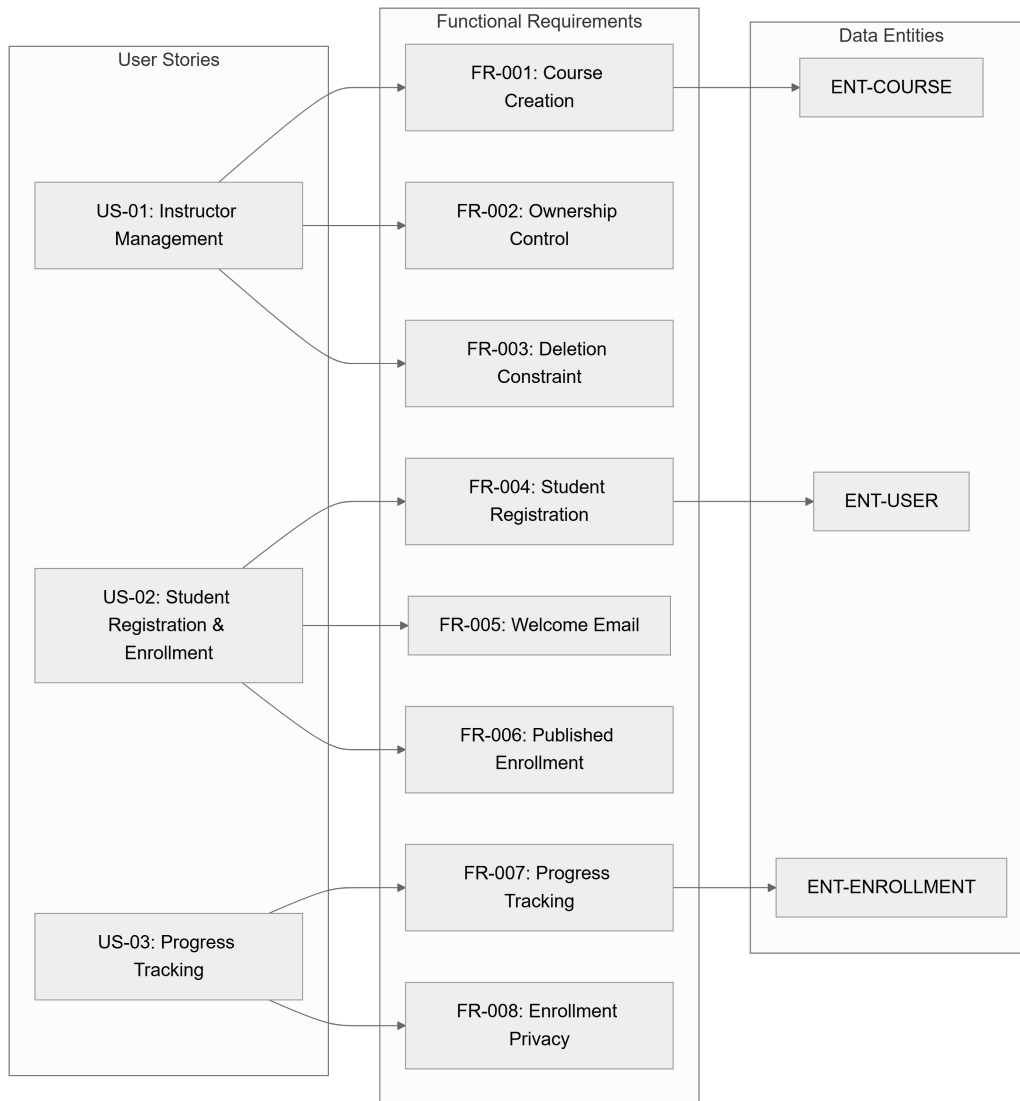
2.3 Course Deletion Workflow

Logic flow for instructors attempting to delete courses, incorporating ownership and active enrollment constraints.



2.4 Requirements Traceability Matrix

Mapping of User Stories to Functional Requirements and their associated Data Entities.



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Type	Description	Related User Story	Related Entity
FR-001	Requirement	Allow instructors to create courses (title, description, price in cents, published flag, ordered modules).	US-01	ENT-COURSE
FR-002	Requirement	Associate each course with a single instructor and restrict management actions to that instructor.	US-01	-

FR-003	Requirement	Reject course deletion when the course has active enrolments and return a clear error message.	US-01	-
FR-004	Requirement	Allow students to register with email and password and assign them the student role.	US-02	ENT-USER
FR-005	Requirement	Send a welcome email through Resend after successful student registration.	US-02	-
FR-006	Requirement	Allow students to enroll only in published courses.	US-02	-
FR-007	Requirement	Track enrolment progress (0-100%) and record completion date at 100%.	US-03	ENT-ENROLLMENT
FR-008	Requirement	Ensure students can view and update only their own enrolments.	US-03	-
FR-009	Requirement	Expose REST endpoints under <code>/api/v1/</code> with JSON envelope <code>{"data": ..., "meta": ..., "errors": []}</code> .	-	-
FR-010	Requirement	Use a shared user table with a role field to distinguish students and instructors.	-	ENT-USER
FR-011	Requirement	Preserve the defined ordering of modules within each course.	-	ENT-MODULE
US-01	User Story	Instructor can create, publish, and manage courses without affecting others.	-	-
US-02	User Story	Student can register, receive welcome email, enroll, and view own records.	-	-
US-03	User Story	Students can update own progress and completion status safely.	-	-

CON-DEL-ENROL	Constraint	Deleting a course with one or more active enrolments must fail with a clear validation error.	-	-
CON-PROG-RANGE	Constraint	Progress values outside the 0–100% range must be rejected.	-	-
CON-PUBLISHED-ONLY	Constraint	Students must not be able to enroll in unpublished courses.	-	-
CON-OWNERSHIP	Constraint	Instructors must not be able to manage courses they do not own.	-	-
SC-001	Success Crit.	Instructor can publish a course and make it available in a single workflow.	-	-
SC-002	Success Crit.	Student can register, enroll, and update progress without accessing others' data.	-	-
SC-003	Success Crit.	>= 90% of core course, enrollment, and authorization behaviors covered by E2E tests.	-	-
SC-004	Success Crit.	System returns a clear, consistent error when deleting a course with active enrolments.	-	-
AS-01	Assumption	Each course has one instructor owner; an instructor may own multiple courses.	-	-

3.2 Security Rules

- **Role-Based Access Control (RBAC)**: Access is governed by the `role` field in the shared User table.
- **Ownership Isolation**: Instructors are strictly forbidden from managing courses they do not own (`CON-OWNERSHIP`).
- **Data Privacy**: Students are restricted to viewing and updating only their own enrollment records (`FR-008`).
- **Authentication**: All management and enrollment endpoints require a valid authenticated session.

3.3 Data Models

Entity	Description	Key Attributes
ENT-USER	Shared account identity	email (PK), password, role, display_name
ENT-COURSE	Learning offering owned by instructor	id (PK), title, description, price_cents, published, instructor_id (FK)
ENT-MODULE	Child record of a course (lesson/section)	id (PK), course_id (FK), sequence_order, content
ENT-ENROLLMENT	Student-course relationship	id (PK), student_id (FK), course_id (FK), progress_percent, completion_date

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Scope & Out-of-Scope	MEDIUM	Explicitly list what the API will NOT handle to avoid scope creep.
Open Questions & Uncertainties	LOW	Document any ambiguities regarding role transitions or email template management.

4.2 Remediation & Workflow

The identified gaps should be addressed during the transition from the "Draft" status to "Final" specification. The Lead Architect must define the boundaries of the API (e.g., whether payment processing is handled by this API or an external gateway) to satisfy the Scope gap.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
API	TECHNICAL_STACK	FR-009	The interface exposing endpoints under /api/v1/ to handle requests and provide responses.
Course	BUSINESS_DOMAIN	ENT-COURSE	A learning offering owned by an instructor, containing a title, description, price in cents, published status, and ordered modules.
Cryptographic Hashing	TECHNICAL_STACK	FR-004	The mechanism used to secure password credentials during registration and authentication.

Enrollment	BUSINESS_DOMAIN	ENT-ENROLLMENT	A relationship between a student and a learning offering that records progress and a completion date.
Fixed-Point Numeric Constraint	TECHNICAL_STACK	FR-001	The requirement to store monetary values in cents to avoid floating point precision errors.
JSON	TECHNICAL_STACK	FR-009	The data interchange format used for responses, following a specific envelope with data, meta, and errors fields.
Module	BUSINESS_DOMAIN	ENT-MODULE	A child record of a learning offering representing a lesson or section in a fixed sequence.
REST	TECHNICAL_STACK	FR-009	The architectural style governing the design of endpoints and resource interactions.
User	BUSINESS_DOMAIN	ENT-USER	A shared account identity containing a role, email, password credential, and display name.